

I PUC MID-TERM EXAMINATION, OCTOBER-2025

Time : 3 Hours

SUBJECT : PHYSICS (33)

Max. Marks : 70

General Instructions :

- 1) All parts A to D are compulsory.
- 2) For Part-A questions, first written answer will be considered for awarding marks.
- 3) Answers without relevant diagram/figure/circuit wherever necessary will not carry any marks.
- 4) Direct answers to the numerical problems without relevant formula and detailed solutions will not carry any marks.

PART-A

I Answer ALL of the following questions:

15x1=15

- 1) The physical quantity has SI unit but not dimensions
a) Radius b) Angle c) Angular velocity d) Angular momentum
- 2) The displacement of a particle is represented by the equation $x = 8t^2 + 3t + 2$ where x is in m and t is in s. What is the acceleration of the particle
a) 8 ms^{-2} b) 4 ms^{-2} c) 16 ms^{-2} d) 3 ms^{-2}
- 3) The slope of v - t graph gives
a) displacement b) velocity c) acceleration d) distance
- 4) The unit vector of vector A is
a) $\hat{n} = |\vec{A}| \vec{A}$ b) $\hat{n} = \frac{|\vec{A}|}{\vec{A}}$ c) $\hat{n} = \vec{A}$ d) $\hat{n} = \frac{\vec{A}}{|\vec{A}|}$
- 5) The component of a vector r along x -axis will have maximum value if
a) r is along positive y -axis b) r is along negative y -axis
c) r is along positive x -axis d) r is along negative x -axis
- 6) For a given angle of projection of the projectile if one initial velocity is doubled. The range of the projectile becomes
a) half b) two times c) one fourth d) four times
- 7) A body executing uniform has at instant its velocity vector and acceleration vector
a) along the same direction b) normal to each other
c) in opposite direction d) not related to each other
- 8) A person jumps from a moving bus, one person falls in
a) backward direction b) Forward direction
c) Remains in straight direction d) None of these
- 9) A Jet engine works on the principle of
a) conservation of mass
b) conservation of energy
c) conservation of Linear momentum
d) conservation of angular momentum
- 10) Which of the following is a self adjusting force
a) Limiting force b) Kinetic friction c) Static friction d) All the three
- 11) The physical quantity which is equal to the change in momentum of a body is known as
a) acceleration b) force c) normal reaction d) Impulse
- 12) Which of the following is not a unit of power
a) Newton b) Watt c) Kilowatt hour d) Horse power

(P.T.O.)



- 13) If the kinetic energy of a body becomes four times of its initial value, then the new momentum will be
 a) Four times of its initial value
 b) Remains same
 c) Three times of its initial value
 d) twice its initial value
- 14) Moment of inertia is independent of its
 a) Nature of distribution of mass
 b) Angular velocity
 c) Axis of rotation
 d) different orientation of body
- 15) The vector product of two vectors is maximum when
 a) $\theta = 45^\circ$
 b) $\theta = 90^\circ$
 c) $\theta = 0^\circ$
 d) $\theta = 180^\circ$

II Fill in the blanks by choosing appropriate answer given in the bracket for all the following questions : 5x1=5
 (decreases, increases, zero, independent, dependent, maximum)

- 16) The vertical component of projectile at maximum height of the projectile is _____.
- 17) The co-efficient of friction _____ with Area.
- 18) In an non conservative forces the work done on a body _____ on the path.
- 19) The potential energy of a body with height _____.
- 20) A diver in a swimming pool bends his head before diving because it _____ his moment of inertia.

PART-B

III Answer any FIVE of the following questions:

5x2=10

- 21) Mention the SI unit of (i) pressure (ii) Torque.
- 22) Define : (i) uniform acceleration (ii) non-uniform acceleration.
- 23) State and explain Triangle law of vector addition.
- 24) Mention the types of Inertia.
- 25) A body of mass 2 kg is placed on a horizontal surface co-efficient of friction 0.5. Calculate the frictional force arises between two surfaces.
- 26) Show that $\mathbf{P} = \mathbf{\bar{F}} \cdot \mathbf{\bar{v}}$ symbols having usual meanings.
- 27) Write the conditions for the mechanical equilibrium of a rigid body.
- 28) Define angular momentum of a particle when it is maximum.

PART-C

IV Answer any FIVE of the following questions:

5x3=15

- 29) Mention the applications of Dimensional analysis.
- 30) The displacement (in metre) of a particle moving along x-axis is given by $x = 18t + 5t^2$. Calculate average velocity between $t = 3s$ and $t = 5s$.
- 31) Derive an expression for Range of the projectile.
- 32) Derive an expression of magnitude of the resultant of two vectors acting at point using parallelogram law of vector addition.
- 33) Derive $\mathbf{\bar{F}} = m\mathbf{\bar{a}}$
- 34) State and prove work-energy theorem for a constant force.
- 35) Give the differences between conservative and non-conservative forces with examples.
- 36) Define vector product of two vectors when it is maximum and minimum.



PART-D

V Answer any THREE of the following questions:

3x5=15

- 37) Derive $V^2 = u^2 + 2as$ using velocity-time graph.
- 38) Deduce an expression for centripetal acceleration.
- 39) State and prove the law of conservation of Linear Momentum using Newton's third law.
- 40) Show that total mechanical energy is conserved for freely falling body.
- 41) a) What is centre of Mass of a body. (1)
b) Write the expression for centre of Mass of system of two particles and explain the terms. (2)
c) Define Radius of gyration. Mention its SI unit. (2)

VI Answer any TWO of the following questions:

2x5=10

- 42) A projectile is fired with horizontal velocity of 30 ms^{-1} from the tops of a cliff 80 m high. How long will it take to strike the level ground at the base of the cliff? With what velocity will it strike? (Neglect air resistance)
- 43) A bullet of mass 0.05 Kg enters a block of wood with a velocity of 300 ms^{-1} and penetrates to a depth of 0.1 m. Calculate the average force of the wood to the penetration of the bullet.
- 44) A train of mass 200 tons is running on level rails with a uniform velocity of 108 Km hr^{-1} . If the resistance to its motion is 2 Kg wt per ton, find power of the engine.
- 45) The speed of a wheel increases from 300 rpm to 900 rpm in 10s. How many revolutions does it make in this interval of time.

